

Newton's Laws

Practice Problems 1–20

Complete worked solutions to the four supplied worksheet photographs. Questions are restated in words so the solutions can be read independently.

Method: Choose a body or subsystem, identify all external forces, choose positive directions, and apply $\Sigma F = ma$. Use the whole system to find acceleration and a smaller subsystem to find tension or contact force.

Conventions: $g = 9.8 \text{ m/s}^2$; cords are ideal and taut unless testing for slack. Connected blocks have equal acceleration magnitudes for these fixed-pulley arrangements. Negligible table friction is assumed in Questions 9, 11, and 17, where no friction data are supplied. Answers generally retain three significant figures for easy checking.

Quick answer key

- 1 A: 4.00 kg; B: 1.00 kg; C: 4.00 kg; D: 1.00 kg
- 2 (a) $T = 41.7 \text{ N}$, $N = 72.1 \text{ N}$; (b) $a = 4.90 \text{ m/s}^2$ down slope
- 3 $F = 566 \text{ N}$; $N = 1.13 \times 10^3 \text{ N}$
- 4 Contacts: 1.23, 2.46, 3.69, 4.92 N; $F = 6.15 \text{ N}$; net/link = 0.250 N up
- 5 $N = 176 \text{ N}$
- 6 (a) 2.18 m/s^2 ; (b) 116 N; (c) 21.0 m/s^2
- 7 $m_2 = 23 \text{ kg}$
- 8 $a = 0.970 \text{ m/s}^2$; $T_1 = 11.6 \text{ N}$; $T_2 = 34.9 \text{ N}$
- 9 $a = 2.50 \text{ m/s}^2$ in both cases; $F = 30.0 \text{ N}$
- 10 $C = 0.123 \text{ N}$; then 3.08 N when pushing the blocks together
- 11 $T_{AB} = 184 \text{ N}$; $T_{BC} = 36.8 \text{ N}$; $d = 0.191 \text{ m}$
- 12 $a = 3.59 \text{ m/s}^2$; $T = 17.4 \text{ N}$
- 13 $a = 0.735 \text{ m/s}^2$; $T = 20.8 \text{ N}$
- 14 (a) 466 N; (b) 527 N, both downward pulls
- 15 $T_{AB} = 68.6 \text{ N}$; $T_{BC} = 81.7 \text{ N}$
- 16 (a) $T = 3.10 \text{ N}$; (b) $F_{\max} = 14.7 \text{ N}$
- 17 $T = 2.40 \text{ N}$
- 18 $T = 4.58 \text{ N}$
- 19 $T = 16.1 \text{ N}$
- 20 $T = 2.60 \text{ N}$; $\beta = 17.2^\circ$

1. Four suspended spheres

Question. Spheres A, B, C, and D hang in a vertical line, from top to bottom. The top cord passes over a frictionless pulley and pulls on the wall with 98 N. The shorter cords have tensions $T_1 = 58.8$ N, $T_2 = 49.0$ N, and $T_3 = 9.8$ N. Find each mass.

1. Set up the forces. The spheres are at rest, so each has zero net force. An upper cord pulls a sphere upward; a lower cord pulls it downward. The tension in the top cord is 98 N.

2. Start at the bottom. D: $T_3 - m_D g = 0$, so $m_D = 9.8/9.8 = 1.00$ kg.

3. Work upward. C: $T_2 - T_3 - m_C g = 0 \rightarrow m_C = (49.0 - 9.8)/9.8 = 4.00$ kg.

B: $T_1 - T_2 - m_B g = 0 \rightarrow m_B = (58.8 - 49.0)/9.8 = 1.00$ kg.

A: $98 - T_1 - m_A g = 0 \rightarrow m_A = (98 - 58.8)/9.8 = 4.00$ kg.

Answer: $m_A = 4.00$ kg; $m_B = 1.00$ kg; $m_C = 4.00$ kg; $m_D = 1.00$ kg.

Why this works / check: Each cord supports the total weight below it. Check: the total mass is 10.0 kg, whose weight is 98 N.

2. Block held on a frictionless incline

Question. An 8.5 kg block rests on a frictionless 30° incline, held by a cord parallel to the incline. (a) Find the tension and normal force. (b) Find its acceleration immediately after the cord is cut.

1. Resolve the weight. Choose axes parallel and perpendicular to the ramp. Gravity contributes $mg \sin \theta$ down the ramp and $mg \cos \theta$ into the ramp.

2. (a) Use equilibrium. Parallel: $T - mg \sin \theta = 0 \rightarrow T = (8.5)(9.8) \sin 30^\circ = 41.65$ N.

Perpendicular: $N - mg \cos \theta = 0 \rightarrow N = (8.5)(9.8) \cos 30^\circ = 72.14$ N.

3. (b) Remove the tension. Once cut, only the weight component acts along the ramp: $mg \sin \theta = ma$.
 $a = g \sin 30^\circ = (9.8)(0.500) = 4.90 \text{ m/s}^2$, down the ramp.

Answer: (a) $T = 41.7$ N; $N = 72.1$ N. (b) $a = 4.90 \text{ m/s}^2$ down the incline.

Why this works / check: On a frictionless incline, the normal force cancels only the perpendicular component of gravity.

3. Horizontal push up a ramp

Question. A 100 kg crate moves at constant speed up a frictionless 30.0° ramp under a horizontal applied force F . Find F and the ramp's normal force.

1. Identify components. Constant speed along the straight ramp means $a = 0$. The horizontal push has component $F \cos \theta$ up the ramp and $F \sin \theta$ into the ramp.

2. Balance forces along the ramp. $F \cos \theta - mg \sin \theta = 0 \rightarrow F = mg \tan \theta$.
 $F = (100)(9.8) \tan 30.0^\circ = 565.80 \text{ N}$.

3. Balance perpendicular forces. $N - mg \cos \theta - F \sin \theta = 0$.
 $N = (100)(9.8) \cos 30.0^\circ + (565.80) \sin 30.0^\circ$
 $N = 848.70 + 282.90 = 1131.61 \text{ N}$.

Answer: $F = 566 \text{ N}$; $N = 1.13 \times 10^3 \text{ N}$.

Why this works / check: The horizontal push presses the crate into the ramp, so N is larger than $mg \cos \theta$.

4. Five accelerating chain links

Question. Five links, each of mass 0.100 kg , accelerate upward at 2.50 m/s^2 . Links are numbered 1 at the bottom through 5 at the top. Find the forces between adjacent links, the applied force, and the net force on each link.

1. Choose a subsystem. For the bottom n links, the upward contact force S_n supports their weight and accelerates them: $S_n - nmg = nma$.
Thus $S_n = nm(g + a) = n(0.100)(9.8 + 2.50) = 1.23n \text{ N}$.

2. Evaluate all four contacts. Between links 1 and 2: 1.23 N . Between links 2 and 3: 2.46 N .
Between links 3 and 4: 3.69 N . Between links 4 and 5: 4.92 N .

At each contact, the upper link pulls up on the lower one; the lower link pulls down on the upper one with equal magnitude.

3. Find the applied and net forces. For all five links: $F = 5m(g + a) = 6.15 \text{ N}$ upward.
For each individual link: $F_{\text{net}} = ma = (0.100)(2.50) = 0.250 \text{ N}$ upward.

Answer: Contact forces, bottom to top: **1.23, 2.46, 3.69, 4.92 N**. Applied force: **6.15 N**. Net force on every link: **0.250 N upward**.

Why this works / check: The lifting force must both balance gravity and produce the upward acceleration; it is not just ma .

5. Box in connected elevator cabs

Question. Cab A (1700 kg) sits above cab B (1300 kg), joined by a cable of tension 1.91×10^4 N. A 12.0 kg box rests on A's floor. Find the normal force on the box.

1. Use the lower cab to find acceleration. Take upward as positive. Cab B has cable tension upward and weight downward:

$$T - m_B g = m_B a.$$

2. Calculate the shared acceleration. $a = (19100 - 1300 \times 9.8)/1300 = 4.8923 \text{ m/s}^2$ upward.

The taut connecting cable makes both cabs share this acceleration.

3. Apply Newton's second law to the box. $N - m_{\text{box}} g = m_{\text{box}} a.$

$$N = 12.0(9.8 + 4.8923) = 176.31 \text{ N}.$$

Answer: $N = 176 \text{ N}$ upward on the box.

Why this works / check: Cab A's mass is unnecessary here: the lower cab already determines the common acceleration.

6. Angled pull and the lift-off threshold

Question. A 5.00 kg block is pulled across a frictionless horizontal floor by a force of 12.0 N at 25.0° above horizontal. (a) Find its acceleration. (b) Find F just before lift-off as F increases. (c) Find the acceleration at that threshold.

1. (a) Use the horizontal component. While the floor supports the block, $a_y = 0$ and $F \cos \theta = m a_x.$

$$a = (12.0 \cos 25.0^\circ)/5.00 = 2.1751 \text{ m/s}^2.$$

Check contact: $N = mg - F \sin \theta = 43.93 \text{ N}$, which is positive.

2. (b) Set the normal force to zero. The floor cannot pull downward. At impending lift-off, $N = 0$:

$$F \sin \theta = mg \rightarrow F = (5.00)(9.8)/\sin 25.0^\circ = 115.94 \text{ N}.$$

3. (c) Recalculate the horizontal acceleration. At the threshold the vertical forces still balance.

$$a = F \cos \theta / m = g \cot \theta = 9.8/\tan 25.0^\circ = 21.016 \text{ m/s}^2.$$

Answer: (a) $a = 2.18 \text{ m/s}^2$. (b) $F = 116 \text{ N}$. (c) $a = 21.0 \text{ m/s}^2$, horizontal.

Why this works / check: Use the increased force from part (b) in part (c), not the original 12.0 N.

7. Four penguins pulled across ice

Question. Four penguins are arranged left to right as 1, 2, 3, 4 on frictionless ice. Given $m_1 = 12$ kg, $m_3 = 15$ kg, $m_4 = 20$ kg, $T_2 = 111$ N between penguins 2 and 3, and $T_4 = 222$ N in the pulling cord, find m_2 .

1. Choose the right-hand pair. For penguins 3 and 4 together, T_4 acts right and T_2 acts left. Internal forces between these two penguins cancel.

$$T_4 - T_2 = (m_3 + m_4)a.$$

2. Find their common acceleration. $a = (222 - 111)/(15 + 20) = 111/35 = 3.1714$ m/s².

3. Use the left-hand pair. T_2 accelerates penguins 1 and 2: $T_2 = (m_1 + m_2)a$.
 $m_2 = T_2/a - m_1 = 111/(111/35) - 12 = 23$ kg.

Answer: $m_2 = 23$ kg.

Why this works / check: Check the whole system: $222/(12 + 23 + 15 + 20) = 3.1714$ m/s².

8. Three connected blocks on a table

Question. Blocks $m_1 = 12.0$ kg, $m_2 = 24.0$ kg, and $m_3 = 31.0$ kg lie left to right on a frictionless table. The rightmost cord pulls with $T_3 = 65.0$ N. Find the acceleration and tensions T_1 and T_2 between the blocks.

1. Treat all blocks as one system. Internal cord tensions cancel. Total mass $M = 12.0 + 24.0 + 31.0 = 67.0$ kg.
 $a = T_3/M = 65.0/67.0 = 0.97015$ m/s² to the right.

2. Isolate the leftmost block. T_1 is the only horizontal force on m_1 :
 $T_1 = m_1 a = (12.0)(0.97015) = 11.6418$ N.

3. Isolate the first two blocks together. T_2 pulls the combined mass $m_1 + m_2$:
 $T_2 = (m_1 + m_2)a = (36.0)(0.97015) = 34.9254$ N.

Answer: $a = 0.970$ m/s² right; $T_1 = 11.6$ N; $T_2 = 34.9$ N.

Why this works / check: A cord farther toward the pulling end must accelerate more mass, so its tension is larger.

9. Two arrangements of contacting blocks

Question. The same horizontal force F_a first pushes block A against B with a 20.0 N contact force. When F_a pushes B against A, A pushes back on B with 10.0 N. Their total mass is 12 kg. Find the acceleration in each case and F_a . Assume negligible table friction.

1. Compare the two systems. The same external force acts on the same total mass, so both arrangements have the same acceleration magnitude a .

2. Use the block without the applied force. First arrangement: B is accelerated by the 20.0 N contact force, so $m_B a = 20.0$.

Second arrangement: B pushes A rightward with 10.0 N by Newton's third law, so $m_A a = 10.0$.

3. Add and solve. $(m_A + m_B)a = 20.0 + 10.0 = 30.0$ N.

$a = 30.0/12 = 2.50$ m/s² in each arrangement.

$F_a = (12)(2.50) = 30.0$ N.

Answer: Both accelerations: 2.50 m/s² right. Applied force: $F_a = 30.0$ N.

Why this works / check: The contact forces differ because the block being accelerated by contact alone differs. The implied masses are $m_A = 4.00$ kg and $m_B = 8.00$ kg.

10. Push the large block or the small block

Question. Two contacting blocks of masses $m_1 = 30.0$ kg and $m_2 = 1.2$ kg rest on a frictionless table. A 3.2 N horizontal push acts on the larger block. Find the contact force, then find it when the same force pushes the smaller block into the larger block.

1. Find the common acceleration. In either arrangement, $a = F/(m_1 + m_2) = 3.2/31.2 = 0.10256$ m/s².

2. Push on the large block. The contact force alone accelerates the small block:

$C = m_2 a = (1.2)(3.2/31.2) = 0.12308$ N.

3. Push on the small block. Now the contact force alone accelerates the large block:

$C = m_1 a = (30.0)(3.2/31.2) = 3.0769$ N.

Answer: Push on large block: $C = 0.123$ N. Push on small block: $C = 3.08$ N.

Why this works / check: The second result assumes the push keeps the blocks together (reverse the push or swap positions). A rightward push on the small block in the original drawing would separate them, giving $C = 0$.

11. A table block with two hanging boxes

Question. Box A (30.0 kg) lies on a horizontal table. A cord runs from A over a frictionless pulley to hanging box B (40.0 kg); a second cord connects B to C (10.0 kg) below it. Released from rest for 0.250 s, find both tensions and the distance traveled. Assume negligible table friction.

1. Write equations along each motion. A moves right; B and C move down. Let T be the A-B cord tension and S the B-C cord tension.

$$A: T = m_A a.$$

$$B: m_B g + S - T = m_B a. \quad C: m_C g - S = m_C a.$$

2. Add to eliminate both tensions. $(m_B + m_C)g = (m_A + m_B + m_C)a$.
 $a = (50.0)(9.8)/80.0 = 6.125 \text{ m/s}^2$.

3. Substitute and use kinematics. $T = (30.0)(6.125) = 183.75 \text{ N}$.

$$S = m_C(g - a) = (10.0)(9.8 - 6.125) = 36.75 \text{ N}.$$

$$d = \frac{1}{2}at^2 = \frac{1}{2}(6.125)(0.250)^2 = 0.191406 \text{ m}.$$

Answer: $T_{AB} = 184 \text{ N}$; $T_{BC} = 36.8 \text{ N}$; each box travels 0.191 m.

Why this works / check: For box B, the lower cord pulls downward. Do not assign the same tension to two different cords.

12. Atwood's machine

Question. Two hanging blocks of masses $m_1 = 1.30 \text{ kg}$ and $m_2 = 2.80 \text{ kg}$ are joined by a massless cord over a frictionless pulley. Find the acceleration magnitude and cord tension.

1. Choose positive along the motion. The heavier m_2 moves down while m_1 moves up. They share the same acceleration magnitude and the same cord tension.

$$m_1: T - m_1 g = m_1 a. \quad m_2: m_2 g - T = m_2 a.$$

2. Add the equations. $(m_2 - m_1)g = (m_1 + m_2)a$.
 $a = [(2.80 - 1.30)(9.8)]/(1.30 + 2.80) = 3.58537 \text{ m/s}^2$.

3. Solve for the tension. $T = m_1(g + a) = (1.30)(9.8 + 3.58537) = 17.40098 \text{ N}$.

Check with m_2 : $T = m_2(g - a)$, giving the same value.

Answer: $a = 3.59 \text{ m/s}^2$; $T = 17.4 \text{ N}$. The 2.80 kg block moves down.

Why this works / check: The net driving force is the difference in weights; the accelerated mass is the sum of both masses.

13. Incline block and hanging block

Question. A 3.70 kg block m_1 on a frictionless 30.0° incline is joined over a massless, frictionless pulley to a hanging 2.30 kg block m_2 . Find the acceleration magnitude and tension.

1. Determine which side wins. Hanging weight: $m_2g = 22.54$ N. Incline component: $m_1g \sin 30^\circ = 18.13$ N. Thus m_2 accelerates down and m_1 accelerates up the incline.

2. Write and add the equations. $m_2g - T = m_2a$; $T - m_1g \sin \theta = m_1a$.

$$a = (m_2g - m_1g \sin \theta) / (m_1 + m_2)$$

$$a = (22.54 - 18.13) / (3.70 + 2.30) = 0.735 \text{ m/s}^2.$$

3. Find the tension. $T = m_2(g - a) = (2.30)(9.8 - 0.735) = 20.8495$ N.

Answer: $a = 0.735 \text{ m/s}^2$; $T = 20.8$ N. The hanging block accelerates down.

Why this works / check: Compare the hanging weight to the incline-parallel weight component, not to the incline block's full weight.

14. Man lifting a bosun's chair

Question. A man and chair have combined mass 95.0 kg. A massless rope passes over a massless, frictionless pulley, with one end attached to the chair and the other held by the man. Find his pull for (a) constant upward velocity and (b) upward acceleration 1.30 m/s^2 .

1. Select the man-plus-chair system. The rope pulls upward at two places: on the chair and on the man's hands. Each force has magnitude T . Gravity acts downward.

Therefore $2T - Mg = Ma$. His downward pull on the rope has magnitude T .

2. (a) Set acceleration to zero. Constant velocity means $a = 0$:

$$2T = Mg \rightarrow T = (95.0)(9.8)/2 = 465.5 \text{ N}.$$

3. (b) Include the upward acceleration. $T = M(g + a)/2 = (95.0)(9.8 + 1.30)/2 = 527.25$ N.

Answer: (a) Pull downward with 466 N. (b) Pull downward with 527 N.

Why this works / check: For the combined man-chair system, there are two upward tension forces. Missing either one doubles the calculated pull.

15. Table block between two hanging masses

Question. A = 6.00 kg and C = 10.0 kg hang on opposite sides of a frictionless table; B = 8.00 kg sits on the table. Separate cords connect A-B and B-C over frictionless pulleys. Find both cord tensions after release.

1. Choose the directions of acceleration. C descends, B moves right, and A rises. All have the same acceleration magnitude.

A: $T_L - m_A g = m_A a$.

B: $T_R - T_L = m_B a$. C: $m_C g - T_R = m_C a$.

2. Add the three equations. $a = (m_C - m_A)g / (m_A + m_B + m_C)$

$a = (10.0 - 6.00)(9.8) / 24.0 = 1.63333 \text{ m/s}^2$.

3. Solve for each tension separately. $T_L = m_A(g + a) = (6.00)(9.8 + 1.63333) = 68.60 \text{ N}$.

$T_R = m_C(g - a) = (10.0)(9.8 - 1.63333) = 81.6667 \text{ N}$.

Answer: Left cord (A-B): 68.6 N. Right cord (B-C): 81.7 N.

Why this works / check: Check on B: $81.6667 - 68.60 = 13.0667 \text{ N} = (8.00)(1.63333)$.

16. Push toward the pulley: slack limit

Question. A 3.0 kg block m_1 on a frictionless horizontal surface is connected over an edge pulley to a 1.0 kg block m_2 on a frictionless 30° downward slope. A horizontal force F pushes m_1 right, toward the pulley. (a) Find T when $F = 2.3 \text{ N}$. (b) Find the largest F before the cord goes slack.

1. Follow the cord constraint. Take rightward for m_1 and down-slope for m_2 as positive. These motions shorten one cord segment and lengthen the other equally.

$m_1: F + T = m_1 a$. $m_2: m_2 g \sin \theta - T = m_2 a$.

2. (a) Add, then substitute. $a = (F + m_2 g \sin \theta) / (m_1 + m_2) = (2.3 + 4.9) / 4.0 = 1.80 \text{ m/s}^2$.

$T = m_1 a - F = (3.0)(1.80) - 2.3 = 3.10 \text{ N}$.

3. (b) Use the zero-tension threshold. At the limiting case, $T = 0$ and the slope block has $a = g \sin \theta = 4.90 \text{ m/s}^2$.

$F_{\max} = m_1 a = (3.0)(4.90) = 14.7 \text{ N}$.

Answer: (a) $T = 3.10 \text{ N}$. (b) $F_{\max} = 14.7 \text{ N}$ ($T = 0$ at the threshold).

Why this works / check: A rope can pull but cannot push. Above this force the taut-cord equations would require negative tension, so the cord becomes slack.

17. Two blocks, two rightward forces

Question. Block A (4.0 kg) and block B (6.0 kg) are connected by a light string. A is left of B. Rightward forces $F_A = 12\text{ N}$ and $F_B = 24\text{ N}$ act on A and B, respectively. Find the tension. Assume negligible horizontal friction.

1. Find the system acceleration. Both applied forces point right. For the combined system, the tension forces cancel:

$$a = (F_A + F_B)/(m_A + m_B) = (12 + 24)/(4.0 + 6.0) = 3.60\text{ m/s}^2.$$

2. Use block A. The string pulls A toward B, to the right:

$$F_A + T = m_A a \rightarrow T = (4.0)(3.60) - 12 = 2.40\text{ N}.$$

3. Check with block B. The string pulls B to the left:

$$F_B - T = m_B a \rightarrow 24 - 2.40 = (6.0)(3.60) = 21.6\text{ N}.$$

Answer: $T = 2.40\text{ N}$.

Why this works / check: The applied 12 N and 24 N forces point in the same direction; tension acts in opposite directions on the two blocks.

18. Pull uphill with a pulley at the bottom

Question. A 12 N force pulls $m_2 = 1.0\text{ kg}$ up a frictionless 37° incline. A cord connects it through a pulley at the bottom to $m_1 = 3.0\text{ kg}$ on a frictionless floor. Find the tension; cord and pulley masses are negligible.

1. Identify the tension direction. On m_2 , the cord pulls toward the bottom pulley, down the incline. On m_1 , it pulls right toward the pulley. Choose uphill for m_2 and rightward for m_1 as positive.

2. Write the equations. m_1 : $T = m_1 a$.

$$m_2: F - T - m_2 g \sin \theta = m_2 a.$$

3. Add and solve. $a = (F - m_2 g \sin \theta)/(m_1 + m_2)$

$$a = [12 - (1.0)(9.8) \sin 37^\circ]/4.0 = 1.52555\text{ m/s}^2.$$

$$T = m_1 a = (3.0)(1.52555) = 4.57666\text{ N}.$$

Answer: $T = 4.58\text{ N}$.

Why this works / check: Tension opposes the applied uphill force here because the pulley is below the incline block.

19. Blocks on opposite slopes

Question. A 3.0 kg block m_1 rests on a frictionless 30° incline and a 2.0 kg block m_2 rests on a frictionless 60° incline. They are connected over a massless, frictionless pulley at the top. Find the cord tension.

1. Compare the downslope weight components. $m_1 g \sin 30^\circ = 14.70 \text{ N}$; $m_2 g \sin 60^\circ = 16.9741 \text{ N}$.
Therefore m_2 moves down its slope and m_1 moves up its slope.

2. Write and add the force equations. $T - m_1 g \sin 30^\circ = m_1 a$; $m_2 g \sin 60^\circ - T = m_2 a$.
 $a = (16.9741 - 14.70)/(3.0 + 2.0) = 0.454820 \text{ m/s}^2$.

3. Find the tension. $T = m_1 g \sin 30^\circ + m_1 a$
 $T = 14.70 + (3.0)(0.454820) = 16.0645 \text{ N}$.

Answer: $T = 16.1 \text{ N}$.

Why this works / check: The lighter block wins here because its slope is steeper. Compare components, not just masses.

20. Find the tension and unknown incline angle

Question. A 1.0 kg block m_1 on a frictionless slope is connected through a pulley at the bottom of the slope to hanging block $m_2 = 2.0 \text{ kg}$. An upward force $F = 6.0 \text{ N}$ acts on m_2 , which accelerates downward at 5.5 m/s^2 . Find T and the incline angle β .

1. Use the hanging block first. Choose downward positive for m_2 . Its weight acts down; tension and the applied force act up:

$$m_2 g - T - F = m_2 a.$$

$$T = m_2 g - F - m_2 a = (2.0)(9.8) - 6.0 - (2.0)(5.5) = 2.60 \text{ N}.$$

2. Follow the actual pulley geometry. As m_2 descends, m_1 moves down the incline toward the bottom pulley. Both tension and $m_1 g \sin \beta$ act down the slope.

$$T + m_1 g \sin \beta = m_1 a.$$

3. Solve for the angle. $\sin \beta = (m_1 a - T)/(m_1 g) = [(1.0)(5.5) - 2.60]/[(1.0)(9.8)] = 0.295918$.
 $\beta = \sin^{-1}(0.295918) = 17.2126^\circ$.

Answer: $T = 2.60 \text{ N}$; $\beta = 17.2^\circ$.

Why this works / check: Because the pulley is at the bottom, tension pulls the incline block downhill. Reversing that force produces the wrong angle.